

CoRoT-2b

R-1245

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_200913 · created 2026-07-31T11:21:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459105.7315 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.061
Transit depth (Rp/Rs)^2	1.92 +/- 1.69 [%]
Orbital Inclination (inc)	89.74 +/- 2.02
Airmass coefficient 1 (a1)	1.168 +/- 0.089
Airmass coefficient 2 (a2)	-0.09 +/- 0.051
Scatter in the residuals of the lightcurve fit is	7.63 %
Best Comparison Star	#3 - [537, 264]
Optimal Aperture	2.03
Optimal Annulus	4.97
Transit Duration (day)	0.0925 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.061 vs 0.1667 (deviation 0.45σ)
Duration fit vs published	0.0925 d vs 0.095 d (deviation 0.41σ)
Mid-time O–C	6.3 min
Depth SNR	2.3
Residual scatter	7.63 %

Light curve

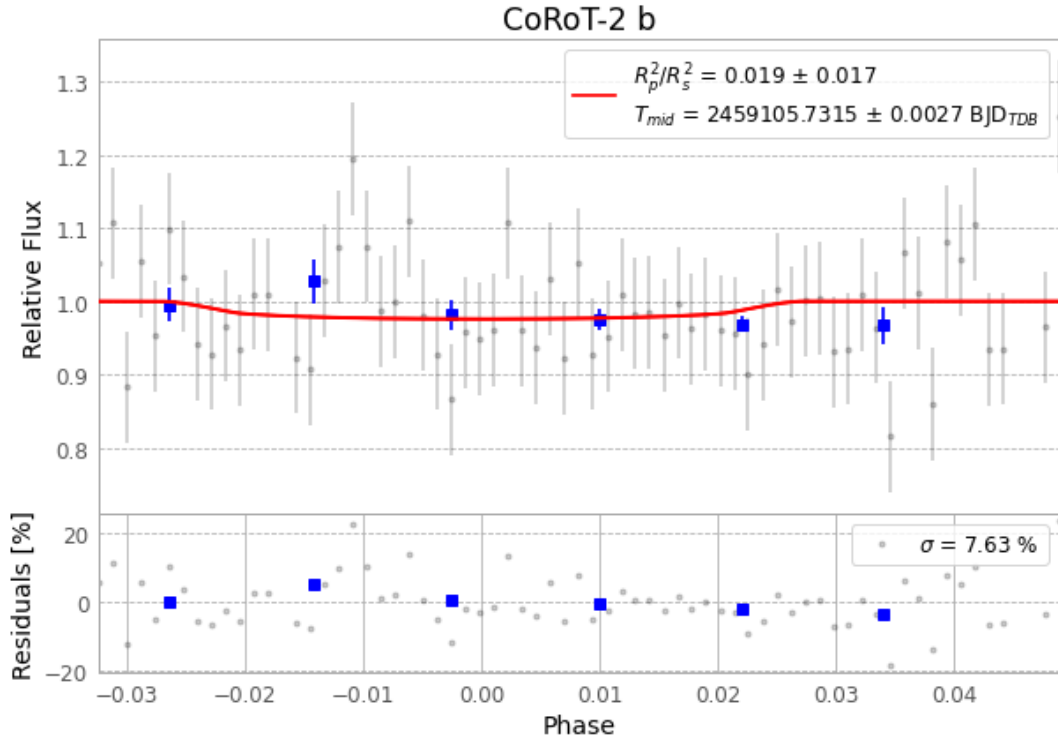


Figure 1 — FinalLightCurve_CoRoT-2 b_2020-09-12.png (SHA-256 a5bdc2e1f827802b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [118, 239], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 f36817cf717a6838...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), CoRoT-2200913040613.FITS → CoRoT-2200913073313.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-200913.log (0374d58b21a6...), FinalLightCurve_CoRoT-2 b_2020-09-12.png (a5bdc2e1f827...), FinalLightCurve_CoRoT-2 b_2020-09-12.csv (cd411195fb6e...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:21Z.